

Google Cloud

Build with AI

Nexus VisBharat

Multilingual civic intelligence
for better public services



People's Voice



Smarter Governance



Stronger Communities

Project Information



Live Demo

nexus-visbharath-510474645723.asia-south1.run.app



Visual Showcase

nexusaitech.in/apps/nvb



Pilot Scope

Rehearsal pilot – Vellore (TN), Tirupati (AP), Bengaluru Urban (KA)



License

Apache 2.0 (Digital Public Good)



Source Code

github.com/sonofmoon/nexus-visbharat



Developer

Dr. Ravikumar Chandrasekaran

Thanthai Periyar E.V. Ramasamy
Government Polytechnic College, Vellore

A submission for Google's Build with AI: [Code for Communities \(2nd Edition\)](#)

02

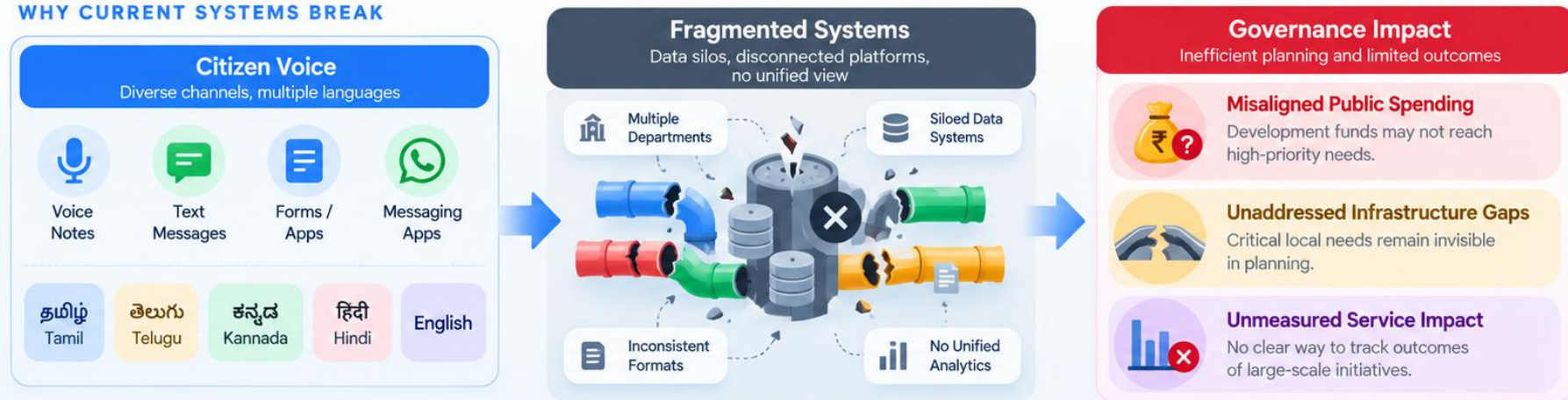
Nexus VisBharat – The Problem

THEME: PROBLEM

FRAGMENTED CITIZEN VOICE

Public-service information often arrives as voice notes, text messages, and local-language descriptions. Valuable citizen signals are difficult to compare across channels, languages, locations, and departments.

WHY CURRENT SYSTEMS BREAK



WHY IT MATTERS



Millions of multilingual requests remain difficult to consolidate.

Citizen signals are spread across channels, languages and regions.



Decision-makers lack a shared operating picture.

It is hard to compare demand, prioritise investments, and align across departments.



Human-led planning becomes slower and less evidence-driven.

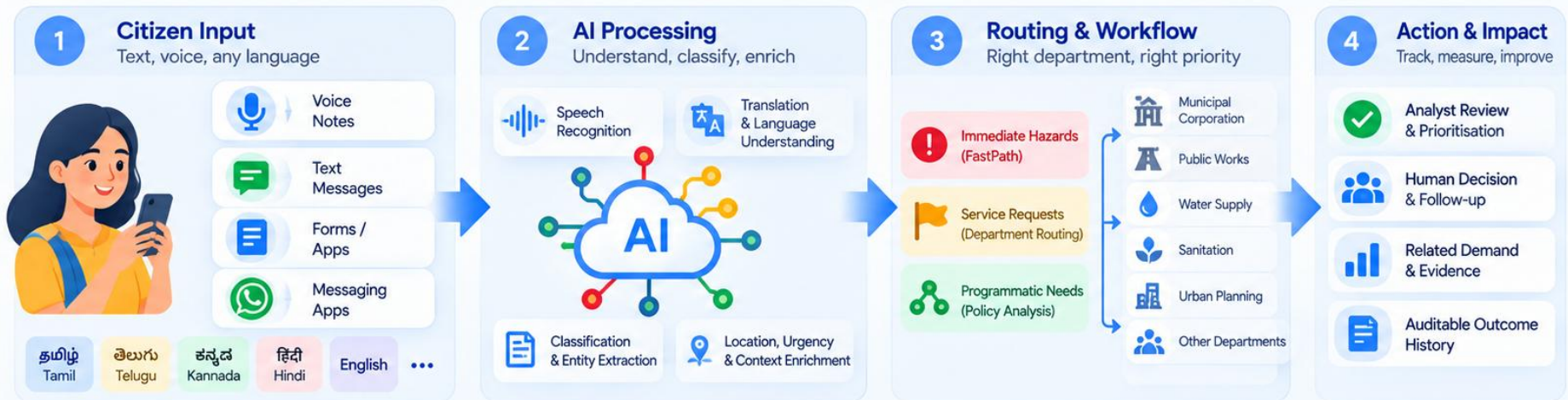
Valuable insights are lost, leading to delayed action and suboptimal outcomes.

03

Nexus VisBharat – The Solution

THEME: INNOVATION

Nexus VisBharat connects citizen voice with the public-service teams responsible for understanding, prioritising and following through on local needs. Citizens can submit a report in text or speech, and the system protects the input, structures the context, routes the case and preserves the human decision trail.



KEY FEATURES



Multilingual by Design

Supports 13 languages (bounded pilot: Tamil, Telugu, Kannada, Hindi, English)



Privacy & Consent

PII scrubbing, role-based access and responsible data handling



Open & Reusable

Apache 2.0 license, containerised deployment and configurable routing



Omnichannel Intake

Web portal, mobile apps, Telegram bot and APIs



Evidence-Driven

Aggregates insights to surface demand hotspots and recommend priorities

04

Nexus VisBharat – Our AI Approach

THEME: INNOVATION



We use **Google AI** to convert multilingual, unstructured citizen inputs into actionable, structured insights by combining speech, language understanding, geospatial intelligence and data analytics — with humans in the loop for final decisions.

AI-POWERED CIVIC INTELLIGENCE PIPELINE



GOOGLE AI & CLOUD STACK USED



Responsible AI



Google AI integrated



13-language inclusive design



Human-in-the-loop governance



Open source (Apache 2.0)



Designed as a Digital Public Good

05

Nexus VisBharat – End-to-End User Journey (How it Works)

THEME: AI IN ACTION – FROM CITIZEN VOICE TO GOVERNANCE INSIGHT

Nexus VisBharat converts **multilingual, unstructured citizen inputs** into **actionable, department-specific insights** using Google AI, with a complete **end-to-end workflow** — from submission to resolution tracking and policy intelligence.

1 Citizen Reports

Voice, text, photo or web form in any Indian language



2 AI Processing

Google AI understands and structures the input



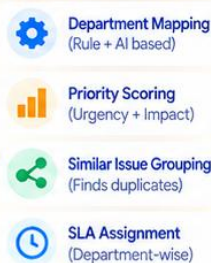
3 Structured Output

Classified, enriched and geo-tagged data



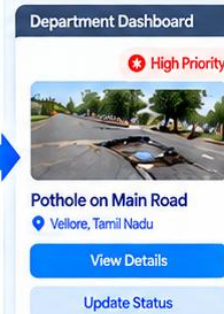
4 Automatic Routing

Sent to the right department with priority and context



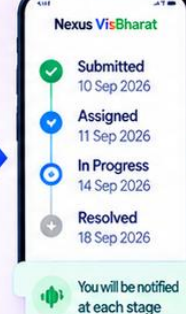
5 Department Action

Officials receive prioritized requests with full context



6 Resolution Tracking

Real-time status updates to citizen



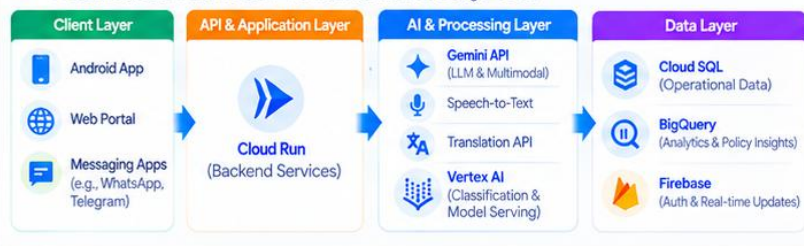
7 Policy Intelligence

Aggregated analytics for better planning and resource allocation



System Architecture

A scalable, secure and modular architecture built on Google Cloud



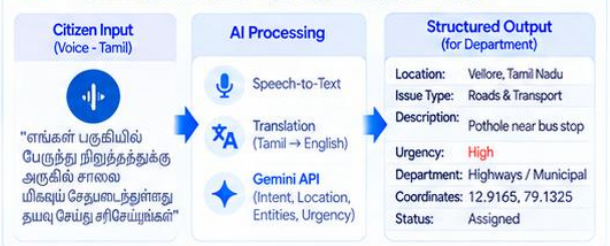
Data Flow & Components

How data moves through the system



Example Transformation (Real Input to Output)

From an unstructured citizen input to structured, actionable data



06

Nexus VisBharat – Pilot Deployment & Use Cases

THEME: IMPACT

We are running a rehearsal pilot in **Vellore (TN)**, **Tirupati (AP)** and **Bengaluru Urban (KA)** to demonstrate how multilingual citizen feedback can be converted into actionable insights for foart local governance.



Pilot Locations

Rehearsal pilot covering three diverse urban districts



Purpose: Validate multilingual intake, local routing, and policy insights before wider rollout.



Key Use Cases

Addressing real civic issues across departments



Roads & Transport

Potholes, damaged roads, street lights, traffic issues



Water Supply

Water leakage, low pressure, no supply, quality issues



Sanitation & Cleanliness

Garbage accumulation, drainage problems, public toilets



Urban Infrastructure

Parks, street infrastructure, public facilities, encroachments



Social & Welfare Services

Community needs, inclusion signals, accessibility issues



Pilot Impact & Learnings

Demonstrating measurable value through a realistic pilot



Faster Issue Intake

Multilingual submissions via web and Telegram



Better Routing

Department-wise automatic routing



Demand Insights

Identify local hotspots and recurring issues



Evidence for Planning

Data-driven insights for better resource allocation



Sample Pilot Issues (Illustrative)

Prepared examples from the pilot environment



Vellore (TN)

Street light not working near bus stop (Tamil voice input)



Tirupati (AP)

Water leakage in residential area (Telugu text input)



Bengaluru Urban (KA)

Garbage collection delay (English web submission)



Real-world scenarios

Based on common civic issues



Configurable for other districts

Reusable pilot framework



Adaptable across states

Supports diverse governance structures



Path to scale

From pilot to state and national rollout

07

Nexus VisBharat – How It Scales Across India

THEME: SCALE

Nexus VisBharat is designed for reuse and scale. The same open source platform, configurable routing and standardized interfaces can be deployed across districts, states and national programmes — enabling a **unified, multilingual civic intelligence layer for India**.

1 District Pilot (Rehearsal)

Validate, learn and refine



- ✓ Real civic issues (prepared examples)
- ✓ Configurable department routing
- ✓ Measure outcomes and iterate

2 State Expansion

Adapt and deploy across departments



- ✓ Reusable deployment framework
- ✓ Configurable district and department mapping
- ✓ Support for state-level governance structures

3 National Platform

A unified civic intelligence layer for India



- ✓ Standardised APIs and data models
- ✓ Scalable cloud-native architecture
- ✓ Enable national programmes and policy insights

ENABLES A STRONGER INDIA



Inclusive Participation

Every voice, every language, every region.



Better Public Services

Data-driven planning and faster issue resolution.



Evidence-Based Policy

Identify national demand trends and infrastructure priorities.



A More Resilient India

Cleaner cities, safer communities and improved quality of life.

KEY ENABLERS FOR SCALE



Modular & Configurable

Adapt to different districts, departments and use cases.



Cloud-Native Infrastructure

Built on Google Cloud (Cloud Run, Cloud SQL, BigQuery, etc.).



Open Standards

REST APIs, JSON schemas and documented interfaces.



Open Source (Apache 2.0)

Reusable, transparent and community-driven.



India-Ready

Multilingual, privacy-conscious and designed as a Digital Public Good.

08

Nexus VisBharat – Impact & Benefits

THEME: IMPACT

Nexus VisBharat creates real impact by turning citizen voice into actionable insights — leading to better public services, more efficient governance, and stronger, more inclusive communities.



Citizens

Easier, Inclusive Participation



- ✓ Report issues in local language (voice or text)
- ✓ Simple and accessible via web, mobile and Telegram
- ✓ Track status and see resolution progress
- ✓ Greater trust and participation



Government Departments

Faster, Smarter Service Delivery



- ✓ Receive well-structured, classified requests
- ✓ Automatic routing to the right department
- ✓ Focus on high-priority and urgent issues
- ✓ Track and manage service levels efficiently



Policymakers & Planners

Data-Driven Decision Making



- ✓ Identify demand hotspots across regions
- ✓ Compare needs with infrastructure indices
- ✓ Plan and prioritise high-impact projects
- ✓ Measure outcomes and long-term impact



Communities & Society

Cleaner, Safer, More Livable Cities



- ✓ Cleaner and safer neighbourhoods
- ✓ Improved infrastructure and civic services
- ✓ More inclusive and equitable development
- ✓ Stronger, more resilient communities

KEY MEASURABLE OUTCOMES (From Pilot & Showcase Environment)



Faster Issue Intake

Multilingual submissions via web and Telegram



Better Routing Accuracy

AI-powered classification and department routing



Clearer Demand Insights

Data-driven analysis across locations and services



Measurable Progress

Track status, resolution time and outcomes



More Inclusive Participation

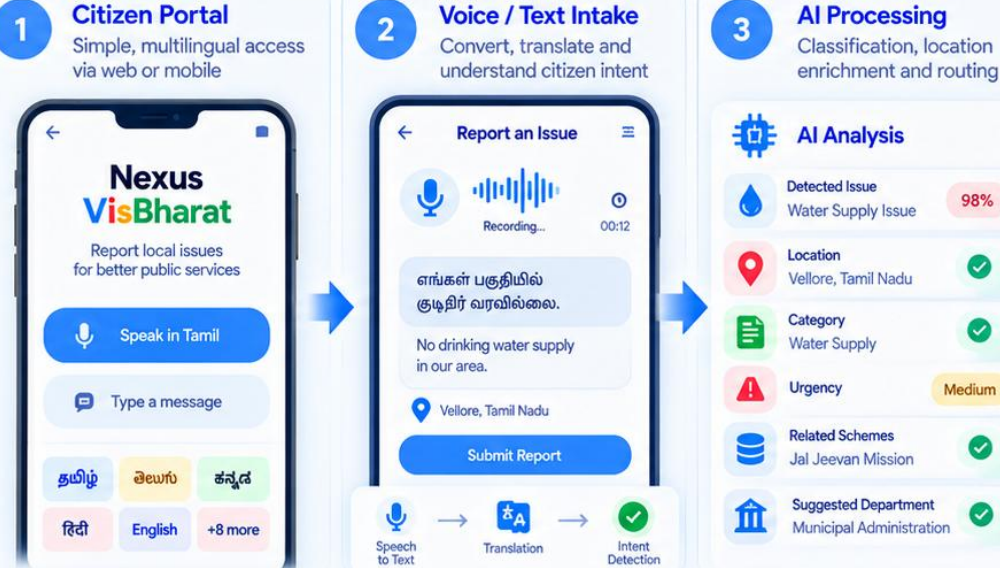
Every voice, every language, every region

09

Nexus VisBharat – Product in Action

THEME: DEMONSTRATION

A working, end-to-end demonstration environment showcasing how citizens can submit issues, how AI processes them, and how government teams can analyse, prioritise and track outcomes.



LIVE DEMO & SHOWCASE



Live Demo

nexus-visbharat-510474645723
asia-south1.run.app



Visual Showcase

nexusaitech.in/apps/nvb



Telegram Bot

@NexusVisBharatBot
Multilingual citizen filing & tracking

KEY HIGHLIGHTS FROM DEMO



Realistic
Citizen Journey



Live Analytics
Dashboard



Map-based
Insights



End-to-End
Tracking

10

Nexus VisBharat – Deployability & Scalability

THEME: DEPLOYMENT



Already built as a working Google Cloud application — designed for real municipal deployment, not just a concept.



DEPLOYMENT ROADMAP

From district pilots to national scale — a reusable, configurable platform.

1

Rehearsal Pilot

Vellore, Tirupati, Bengaluru Urban



- ✓ Validate, learn and refine
- ✓ Real civic issues
- ✓ Measure outcomes

2

Ministry Pilot

Deploy within a ministry or central programme



- ✓ Integrate with central departments
- ✓ Configurable department routing
- ✓ Role-based access and governance

3

State Rollout

Adapt and deploy across districts and departments



- ✓ Reusable deployment framework
- ✓ Standardized APIs and data models
- ✓ Support for state governance structures

4

National Scale

A unified civic intelligence layer for India



- ✓ Scalable cloud-native architecture
- ✓ Stausable across programmes
- ✓ Enable national policy insights



Ready in Weeks

Designed for quick and practical deployment within ministries or across states.

- ✓ Live Cloud Run deployment
- ✓ Containerized architecture
- ✓ Configurable department routing
- ✓ REST APIs
- ✓ Cloud SQL for operational records
- ✓ Documented deployment runbook



Built for Rapid Adoption

Leveraging Google Cloud, open standards and modular design for easy adoption across India.



Google Cloud (Cloud Run)

Scalable, secure and production-ready



Open Source (Apache 2.0)

Transparent, reusable and community-driven



Multilingual by Design

13-language architecture for inclusive participation



Role-Based Access & Audit Trail

Secure, accountable and traceable governance



Reusable APIs & Configurable Routing

Adaptable to different departments and use cases

11

Nexus VisBharat – Why It Matters

THEME: DIGITAL PUBLIC GOOD

Every citizen voice deserves to shape public services.

From fragmented complaints to measurable civic intelligence — built for India, powered by Google Cloud, and designed as a Digital Public Good.



Problem–Solution Fit

Addresses real civic issues faced by citizens



AI / Technical Execution

Speech-to-Text Translation (13 languages)
Gemini for classification



Depth & Reach Across India

Scalable from one city to communities across India



Impact Potential

Evidence-based decisions for cleaner, safer, more liveable cities



Deployability & Scalability

Live Google Cloud application, ready for pilots and national scale



THE VISION



One Platform

Unified civic intelligence for India.



Every Language

Inclusive participation across regions.



Every District

Reusable, configurable deployment.



Building a Digital Public Good for a Cleaner, Safer, and More Inclusive India

Powered by Google Cloud | Open Source | Responsible AI | People-Centric Governance



12

Nexus VisBharat – Thank You

BUILDING INDIA'S CIVIC INTELLIGENCE LAYER

Together
for a Better
India



Dr. Ravikumar Chandrasekaran

Lecturer (Physics)

Thanthai Periyar E.V. Ramasamy
Government Polytechnic College, Vellore

Thank You!

**Every Citizen Voice.
A Cleaner, Safer, More Inclusive India.**

Turning multilingual feedback into measurable
public-service outcomes.



**Powered by
Google Cloud**

Production-ready
deployment



**Multilingual
by Design**

13 languages
for India



**Digital
Public Good**

Open source
foundation



**Evidence-Based
Impact**

Better public decisions
for cleaner, safer cities



**Live Today.
Ready to Scale Tomorrow.**



Live Demo



Visual Showcase



Telegram Bot



Open Source



Built with Google Cloud | Open Source | Human-in-the-Loop Governance